

Impact of Preprocessing, Model Complexity, and Regularisation on Seizure Prediction Generalisation

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Abstract— This paper investigates how preprocessing pipeline design, regularisation strategy, and class imbalance handling affect generalisation performance in epileptic seizure prediction tasks. Using three distinct EEG-based datasets (UCI Epileptic Seizure Recognition, EEG Eye State, and a synthetic CHB-MIT simulation), we evaluate logistic regression under L1, L2, and Elastic Net regularisation. We demonstrate that the ordering of preprocessing steps significantly impacts model performance, that Elastic Net provides the most consistent generalisation across datasets, and that class weighting combined with L2 regularisation achieves optimal precision-recall tradeoffs for the highly imbalanced clinical scenario. Our findings provide practical guidance for designing robust seizure detection pipelines.

Keywords— Epileptic seizure prediction, EEG classification, regularisation, preprocessing pipeline, class imbalance, logistic regression, SMOTE

I. INTRODUCTION

Epilepsy affects approximately 50 million people worldwide, with nearly 30% of patients being drug-resistant [1]. Automated seizure detection from electroencephalography (EEG) offers a promising path toward real-time monitoring and responsive neurostimulation [2]. However, the success of machine learning models in clinical practice depends critically on decisions made long before model training — namely, how raw EEG signals are preprocessed, how model complexity is controlled, and how the chronic class imbalance between seizure and non-seizure epochs is handled [3].

This work makes three primary contributions: (1) a systematic comparison of two preprocessing pipelines and their ordering effects on downstream performance; (2) a controlled overfitting/underfitting study with learning curve analysis; and (3) a comparative regularisation study spanning L1, L2, and Elastic Net penalties across three datasets of varying dimensionality, size, and imbalance ratio [4].

II. DATASETS

A. Dataset 1: UCI Epileptic Seizure Recognition

Source: UCI ML Repository (#388) [5]. This dataset contains 11,500 observations derived from 500 EEG recordings of both epileptic and non-epileptic subjects. Each sample comprises 178 features representing EEG amplitude measurements over a 1-second epoch sampled at 178 Hz. The original five-class label is binarised (class 1 = seizure; classes 2–5 = non-seizure), yielding approximately 20% positive class rate.

C. Ordering Effect Experiment

To empirically validate the ordering hypothesis, we compared Pipeline A (Scale → Select) against an alternative ordering (Select → Scale). Results confirmed that the correct ordering (normalise first) consistently achieved higher PR-AUC, with improvements of 0.02–0.08 across datasets. This validates the theoretical prediction that magnitude-unaware feature selection produces suboptimal feature subsets [9].

IV. BASELINE MODEL AND METRICS

Logistic Regression was selected as the baseline model for its interpretability and direct connection to regularisation theory [4]. The decision boundary is given by:

$$P(y=1|x) = 1 / (1 + \exp(-(\beta_0 + \beta^T x)))$$

For class-imbalanced data, PR-AUC is prioritised over accuracy, as accuracy can be trivially high (e.g., 90% by predicting all non-seizure on D3) [11]. The F1-score balances precision and recall, while PR-AUC provides a threshold-independent summary of the precision-recall tradeoff.

V. OVERFITTING AND UNDERFITTING STUDY

Three scenarios were constructed on Dataset 1:

Underfitting: $C = 0.00001$ (extreme regularisation) with only 3 features. Both training and validation F1 remain below 0.35, indicating the model cannot capture the underlying signal.

Overfitting: $C = 10,000$ (near-zero regularisation) with all 178 original features. Training F1 reaches 0.95+ while validation F1 drops, revealing overfitting.

B. Dataset 2: EEG Eye State

Source: UCI ML Repository (#264) [6]. This dataset contains 14,980 instances with 14 continuous EEG features recorded via a consumer-grade Emotiv headset. Although originally an eye-state classification task, it serves as a lower-dimensional EEG binary classification benchmark with mild class imbalance (~55:45), useful for assessing regulariser stability under minimal distributional shift.

C. Dataset 3: Synthetic CHB-MIT Simulation

Source: Synthetically generated using `sklearn.datasets.make_classification` to simulate the CHB-MIT Scalp EEG Database's [7] severe class imbalance. This dataset contains 5,000 instances with 50 features, 15 informative, 10 redundant, and a 90:10 non-seizure:seizure split. It is used specifically to stress-test imbalance handling techniques.

TABLE I: DATASET SUMMARY

Dataset	Source	Samples	Feat.	Ratio (0:1)
D1 Epileptic	UCI #388	11,500	178	80:20
D2 Eye State	UCI #264	14,980	14	55:45
D3 CHB Sim	Synthetic	5,000	50	90:10

III. PREPROCESSING PIPELINES

Two distinct preprocessing pipelines were designed to assess the impact of step ordering on model generalisation [8].

A. Pipeline A: Normalization → Noise Removal → Feature Selection

Pipeline A first applies `MinMaxScaler` to normalise all features to $[0, 1]$, ensuring that the subsequent `VarianceThreshold` step removes genuinely low-information features rather than simply low-magnitude ones. `SelectKBest` with ANOVA F-statistic is applied last to retain the k most discriminative features. This ordering prevents magnitude bias from contaminating feature selection decisions [9].

B. Pipeline B: Standardisation → PCA

Pipeline B applies `StandardScaler` before PCA, a crucial ordering since PCA identifies directions of maximum variance and will be dominated by high-scale features if standardisation is omitted [10]. PCA retains components explaining 95% of total variance, achieving dimensionality reduction while preserving information for linear separability.

Balanced: $C = 1.0$ with 30 selected features. Training and validation F1 converge and stabilise, indicating effective generalisation [12].

Learning curves confirm these patterns: the underfitting scenario shows flat, low curves regardless of training size; the overfitting scenario shows a persistent gap between training and validation curves; and the balanced scenario shows curves that converge as training size increases [13].

VI. REGULARISATION STUDY

The regularised logistic regression objective is:

$$J(W, b) = (1/m) \sum L(y^{(i)}, y^{(i)}) + (\lambda/2m) \sum \|W\|^2$$

Three regularisation penalties were evaluated: L1 (Lasso), L2 (Ridge), and Elastic Net ($\alpha \times L1 + (1-\alpha) \times L2$, with $\alpha = 0.5$) [14]. Sparsity analysis revealed that L1 zeros out the most coefficients (highest sparsity), performing implicit feature selection [15]. L2 distributes regularisation across all weights, maintaining denser coefficient vectors. Elastic Net achieves a balance between sparsity and stability [16].

TABLE II: REGULARISATION COMPARISON (PR-AUC, C=1.0)

Regularizer	D1 AUC	D2 AUC	D3 AUC
L1 (Lasso)	0.782	0.741	0.623
L2 (Ridge)	0.798	0.763	0.651
Elastic Net	0.804	0.758	0.668

VII. CLASS IMBALANCE HANDLING

Three imbalance mitigation techniques were evaluated on Dataset 3 (90:10 ratio):

SMOTE [17]: Generates synthetic seizure samples by interpolating between existing minority-class instances. Increased recall significantly ($0.41 \rightarrow 0.73$) at the cost of moderate precision decrease.

Random Undersampling [18]: Reduces majority-class samples to match minority count. Most aggressive recall improvement (0.78) but introduces information loss and highest precision drop.

Class Weighting [19]: Penalises misclassification of minority samples more heavily during training. Achieves the best precision-recall balance, making it the preferred technique for clinical deployment where false positives also carry cost.

VIII. COMPARATIVE ANALYSIS AND FINDINGS

Finding 1 — Preprocessing order significantly affects performance. Applying normalisation before feature selection yielded consistently higher PR-AUC (mean improvement: +0.034) compared to the reverse order [8], [9]. Scaling ensures that feature importance scores are computed in a magnitude-normalised space.

Finding 2 — Elastic Net generalises best across datasets. While L2 is competitive on individual datasets, Elastic Net achieves the lowest performance variance across D1, D2, and D3, making it the most robust choice for deployment across patient populations and recording conditions [16].

Finding 3 — Elastic Net does not always outperform L1/L2 individually. On D2 (low-dimensional, balanced), L2 slightly outperforms Elastic Net. On D3 (high imbalance), L1+SMOTE achieves the highest F1. Context-dependent selection remains important [14], [15].

Finding 4 — Class weighting synergises best with L2 and Elastic Net regularisation. L1's sparsification can interact negatively with class weighting on small minority-class feature subsets, occasionally collapsing recall to zero on D3. Class weighting with L2 provided the most stable clinical performance profile [19].

IX. CONCLUSION

This study demonstrates that generalisation performance in seizure prediction is highly sensitive to preprocessing design decisions, not just model selection. The correct ordering of normalisation, noise removal, and feature selection is a prerequisite for reliable performance [8]. Among regularisation strategies, Elastic Net offers the best cross-dataset robustness [16], while class weighting provides the most clinically appropriate imbalance correction [19]. Future work should extend this analysis to deep learning architectures (LSTMs, EEGNet) [20] and real-time streaming scenarios [2].

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